

Full Instructor Guide

EM Evidence Sim · updated 1 October 2026

This guide explains everything an instructor needs to set up, run and debrief a scenario. New to the app? Start with the Quick Start Guide.

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At a glance

A simulation uses up to three screens, all linked online by one six-character session code.

One session code links every screen

Changes you make reach the team's screens; every press on the defib comes back to you



How the screens link: three devices, one session code.

You drive the patient from the instructor screen. The room monitor shows the team only what they have attached. The defib tablet, if used, sends every press back to your event log.

Before the session

Allow 10 minutes to set up the screens and test them before the team arrives.

Equipment

Device	Role	Where it goes
Laptop or tablet	Instructor screen: you drive the patient from here	Out of the team's sight, ideally where you can see the room
Second screen, TV, projector or device	Room monitor: the patient monitor the team watches	Where a real monitor would be, visible to the whole team
Tablet (optional)	Defibrillator	At the bedside, for Defib Sim or any scenario that needs a defib
Speakers (optional)	Alarms and monitor beeps	Next to the room monitor

All devices need an internet connection, and any up-to-date web browser. You do not need an account to run the built-in scenarios.

Pre-flight checklist

- ☐ Open wmebemsim.netlify.app on the instructor laptop and note the session code
- ☐ Open the room monitor and tap **Tap to Enable Sound**
- ☐ Link the defib tablet, if you are using one
- ☐ Check the badge on the instructor screen reads **Live** and names the linked screens
- ☐ Load your scenario and read its brief
- ☐ Turn the room monitor's volume up and check you can hear an alarm
- ☐ Change one vital sign and check the room monitor follows

Session codes and linking screens

The six-character session code (for example K7PQ3M) links every screen in one simulation. It never uses look-alike characters (no 0/O or 1/I/L), so it is easy to read out. It shows on the setup screen and as "Session ID" in the top right corner.

- **Same code all day:** one code can run any number of scenarios with the same group. The linked screens stay linked between scenarios.
- **New code:** press **New code** on the setup screen for a new group. Screens on the old code must join again.

Ways to link a screen

Screen	On the same laptop	On a separate device
Room monitor	Launch Monitor (setup screen) or Screens → Launch room monitor . Opens a new window to drag onto the second screen	Scan the QR code from Screens → Join by QR code , or open the site, press Join Another Session , type the code, press Connect
Defib tablet	Screens → Open defib tablet (not offered in Quick Sim). A defib opened in another tab of the same browser links without a code	Scan the defibrillator QR code from Screens → Join by QR code , or open wnebensim.netlify.app/defib/ and type the code

Checking the link: the badge in the instructor top bar shows the connection.

Badge	Meaning
Live · no screens linked	Your session is online but nothing has joined yet
Live · patient monitor (or "2 screens linked")	Linked and working
Syncing	Connecting; wait a few seconds
Sync error	The connection has failed; check the internet

Hover over the badge to see each linked screen and when it was last heard from.

Choosing a scenario

The setup screen has a tab for each way to start. Premade covers most teaching; Defib Sim is for defibrillator skills.

Tab	What you get	Best for
Premade	254 written scenarios in 8 categories: Adult Medical, Trauma, Paediatric, Cardiac Arrest, Toxicology, Obs & Gynae, Elderly, Psychiatric	Planned teaching with set objectives
Quick Sim	A blank patient: you set age, weight, sex, name and starting rhythm. No scenario, drugs or scoring	Ad-hoc teaching, rhythm recognition, obs drills
Defib Sim	Defibrillation, cardioversion and pacing on the defib tablet, with its own controller	Defib skills and assessment (see Running a Defib Sim)
Random	A generated patient from your choice of category, age group and acuity	Unpredictable cases for experienced teams

Tab	What you get	Best for
Custom	Scenarios you have saved or imported as a file	Your own material
Builder	A form to write a scenario field by field	Writing new scenarios
Restricted	Copyright-restricted scenarios (for example RCUK)	Only if your account has been given access

Human factors. Random and Builder can add a challenge: Blindfolded Lead, Silent Team, New Junior, Missing Kit or Distracted Senior. It shows as a badge on your screen so you can brief the right person.

Paediatric scenarios. WETFLAG starts hidden on the room monitor. Tick **Show WETFLAG on Monitor from the start** on the setup screen if you want the team to see it straight away. You can show or hide it at any time during the scenario (see Paediatric scenarios).

The brief. Loading a Premade, Random or Custom scenario opens a brief before anything starts:

- **Patient Brief:** name, history, medicines and allergies
- **Required Equipment**
- **Clinical Progression:** how the scenario is expected to unfold
- **Key Interventions** the team should perform
- **Guidelines & Resources**
- **Learning Objectives:** what the debrief scores against

Press **Start Scenario** to begin, or **Back to Setup** to choose another.

Resume Previous? If the instructor tab was closed during a scenario, the setup screen offers to resume it. The app saves your progress every few seconds.

A tour of the instructor screen

The screen has a top bar, the patient on the left, and treatments and the event log on the right. Drag the divider between the columns to widen either side.

Top bar

Control	What it does
Back	Returns to the setup screen
Finish	Ends the scenario and opens the debrief (asks you to confirm)
PAUSE / START	Stops or restarts the clock
Connection badge	Shows the session is live and which screens are linked

Control	What it does
Safety flags	Appears when the team does something out of the usual order; opens the list
Muted	Appears in red while alarms are muted; press to unmute
Simple / Full	Hides or shows the advanced controls (below)
Screens ▼	Launch room monitor, Join by QR code, Open defib tablet
Tools ▼	Drug calculator, Timer alerts, Full simulation log, where sound plays, Mute alarms, Keyboard shortcuts

Left column: the patient

Each grey bar is a section. Click its title to open or close it; a closed section shows a one-line summary. The app remembers which sections you keep open on that device.

Section	Starts	Contains
Patient details	Always shown	Name, age, scenario, diagnosis, story, any human factors challenge
Monitoring & access	Closed	What is attached (ECG, sats, cuff, temperature, capnography, arterial line, IV access); Attach stays on the bar
Monitor strip and vitals	Always shown	The waveforms, Show timer , Change rhythm , and tiles for HR, BP, SpO2, RR, Temp and ETCO2
More obs	Open	GCS and glucose (plus pH and K+ in Full view)
Drugs on board	Open, once a drug is given	Each drug's phase, strength and time left
Patient condition	Closed	AUTO or MANUAL deterioration; Better and Worse stay on the bar
Rhythm & resus	Open	ARREST, ROSC, Arrest View, Defib, Cycle NIBP Now, Auto NIBP, Manual Reading, Show/Hide WETFLAG on Monitor (child scenarios), and the rhythm and shock tally

Right column: treatments and the log

- **Search** finds any treatment by name or route (for example "IM").
- **Tabs:** Common, Drugs, Airway, Breathing, Circulation, Procedures, Investigations, Voice and Assessment. In Common, the scenario's **Recommended actions** sit at the top in an amber box.
- **Voice:** type what the patient says and press **Speak**; it is read aloud on whichever screen plays sound.

- **Assessment:** tick or cross Safe Approach, Team Leadership, Communication, CPR Quality, Defib Safety and Re-evaluation as you watch.
- **Event log** (bottom): always on screen; see The event log, notes and flags.

Simple and Full view

Simple is the default and is remembered on each device. It hides:

- the drug-timing detail lines in Drugs on board
- the "+ Invasive" button that attaches IV access, an arterial line and capnography in one go (the separate buttons stay)
- the pH and K⁺ tiles
- the defib's shock-response settings and "Next shock converts to" (one line shows what is set; click it to switch to Full)
- the Quick Sim preset list (a preset that is running stays on screen)

Nothing is removed in either view.

On a phone, the obs tiles, rhythm, clock and START/PAUSE sit at the top, and the page scrolls.

Running the patient

You mirror what the team does: they act in the room, you press the matching control, and the patient and room monitor respond.

Monitoring and access

The room monitor only shows what the team has attached. Open Monitoring & access to see a chip for each item.

- Click a chip to attach it (○) or remove it (●). Removing one blanks its trace and number on the room monitor. Every change is logged.
- **Attach** puts on the standard four: ECG, SpO₂ probe, NIBP cuff and temperature probe. Capnography, the arterial line and IV access are separate, deliberate steps.
- **Remove all** takes every monitor off. IV access stays, because it is a route, not a monitor.
- **POC glucose** and **POC VBG** take a one-off reading at that moment. Click again to take a new one.

Vital signs

Your tiles always show the patient's true values, even when the team cannot see them; "not on monitor" means the team cannot see that number yet.

1. Click a vital's tile.
2. Type the new value. For ETCO₂ you can also pick the capnogram shape: Auto, Obstructive, Force normal, Rebreathing or Curare cleft (see Capnography).

3. Under **Get there**, choose **Now, 30 s, 2 min** or **5 min** for a gradual change. BP starts on **Now**; everything else starts on 30 s.
4. Press **Confirm**. For BP there are two buttons instead (see below).

A heart rate you set yourself is kept if you later change the rhythm. ARREST and ROSC reset it.

Blood pressure

Changing the BP never sends it to the team by itself. The room monitor and the defib tablet keep showing the last cuff reading until the cuff cycles, as in real life. The cuff cycles when the team presses **Cycle** on the monitor or **NIBP** on the defib, when you press **Cycle NIBP Now**, or every 3 minutes while auto is on.

When you set a BP with **Now** and the cuff is on, you choose:

- **Change, don't send**: the patient's BP changes now. The team sees it the next time the cuff cycles.
- **Change and cycle cuff now**: the BP changes and the cuff starts measuring straight away. The cuff noise plays and the new reading appears after about 5 seconds.

If you pick 30 s, 2 min or 5 min instead, the BP drifts to the new value and the team sees wherever it has got to at the next cuff cycle.

Auto NIBP: on/off, next to Cycle NIBP Now, is the same switch as **Auto** on the room monitor: the cuff measures every 3 minutes. It also counts down in Quick Sim, where you never press START. An arterial line, once attached, shows the BP continuously. **Manual Reading** shows a one-off cuff reading on the room monitor without changing the patient (for example a spurious reading).

Capnography

Once capnography is attached, the room monitor draws the capnogram at half the speed of the ECG, as the ZOLL R Series does. Several breaths fit on screen, and each one is wide enough to read its shape. The height of the plateau always matches the ETCO2 number.

Shape	When you see it
Normal: a sharp rise, a flat plateau ending at the ETCO2, then a sharp drop to zero as the patient breathes in	Normal breathing or ventilation. A slow rate gives wide waves with a long plateau; a fast rate gives narrow ones
Obstructive ("shark fin"): a slow, sloping rise with no flat plateau	Bronchospasm. On Auto it follows the scenario's bronchospasm and softens as bronchodilators work; Obstructive forces it
Rebreathing: the trace does not return to zero between breaths	Choose Rebreathing
Curare cleft: a notch in the plateau	Choose Curare cleft

Shape	When you see it
No waveform: a flat line at zero	Shows by itself when nothing is moving gas: apnoea, an oesophageal tube or a disconnected circuit
Low trace with small ripples from chest compressions, a breath every 6 seconds	During CPR (10 ventilations a minute)

The ETCO2 control (tap the ETCO2 tile) chooses the shape: Auto, Obstructive, Force normal, Rebreathing or Curare cleft.

Treatments and drugs

Find the treatment by search or tab and click it. The app then:

- logs it with the time
- applies its effect over a realistic time course: onset, peak and wearing off
- shows it in **Drugs on board** with its phase (for example ONSET), strength and time left
- for infusions and oxygen, keeps it running until you click it again; use + and – in Drugs on board to change an infusion's rate

If the team gives a drug before its prerequisite (for example an IV drug with no IV access), the app records a **safety flag**. Child doses follow the RCUK paediatric guidance (see Paediatric scenarios).

Investigations

The team can tap **12-Lead**, **VBG**, **CXR**, **Urine**, **POCUS** or **CT Head** on the room monitor to see results. From the **Investigations** tab you can also send a result yourself: the scenario's own finding, a quick-select finding or one you type.

Deterioration

Patient condition shows the mode:

- **AUTO**: the patient gets worse on their own, at the rate the scenario sets, unless the team treats them.
- **MANUAL**: the obs only change when you change them.

Better and **Worse** move the patient towards the scenario's improved or worsened obs over 30 seconds, in either mode. In many scenarios, giving one of the key treatments starts the improvement by itself, and the log says "Patient condition IMPROVING".

Quick Sim has no scenario, so it leaves out treatments, drugs, deterioration and scoring. Its right side holds the full rhythm list, presets (Full view), Better and Worse, and the event log.

Arrests, rhythms and the defibrillator

All rhythm and resuscitation controls are in the **Rhythm & resus** section.

Control	What it does
ARREST ▼	Pick the arrest rhythm. The obs drop to arrest values and the arrest timer starts
ROSC ▼	Pick the post-ROSC rhythm. Circulation returns
Change rhythm (above the vitals)	Change to any rhythm without an arrest or ROSC, for example sinus to AF
Arrest View	Switches the room monitor to the arrest layout: Lead II, pads and, when capnography is attached, the capnogram with the ETCO2. It does not open a defib
Defib	Opens a working defibrillator on the room monitor, with the obs still visible, and a defib panel on your screen

The panel under these buttons shows the current rhythm (marked shockable, sync cardiovert or pulseless), the number of shocks, total and last energy, and whether CPR is running.

Using the defib panel on your screen: choose the energy (the recommended energy is marked), then **Charge** and **SHOCK**. **SYNC** switches synchronised cardioversion on and off. **CPR start/stop** and **Rhythm check +2m** (jump forward two minutes) are there too, with the CPR cycle timer.

What a shock does. By default a realistic model decides: the energy, the rhythm, whether CPR is running and the drugs given all count. In Full view you can override it:

- **Next shock converts to:** the next shock gives exactly the rhythm you pick.
- **Shock response:** converts on the 1st to 5th adequate shock, never (refractory), Auto or the realistic model.
- **Unsync shock with a pulse:** whether an unsynchronised shock into a perfusing rhythm causes VF (never, sometimes or always).
- **After ROSC:** whether VF comes back.
- **Cardioversion succeeds at:** the lowest energy at which cardioversion works.

When a shock converts the rhythm, or fails to, a pop-up tells you on your screen only. The team never sees it.

Learner defib on a tablet. For defib skills with a real-looking device, use the defib tablet (see Running a Defib Sim). It works in a normal scenario too: link it with **Screens → Open defib tablet** or the QR code.

Running a Defib Sim

In a Defib Sim the learner works a ZOLL-style defibrillator on a tablet, and you run the patient from the **Defib controller**, which replaces the normal instructor screen.

Setting it up

1. On the setup screen, choose the **Defib Sim** tab.
2. Choose the mode:
 - **Education:** the defib shows pulse-check results and RCUK hint cards.
 - **Assessment:** no pulse-check results or hint cards, as on a real defib. Feedback comes in the debrief.
3. Choose a scenario: Ventricular Fibrillation, Pulseless VT, VT with a pulse (unstable), SVT with life-threatening features, Fast AF with life-threatening features, Complete heart block, Symptomatic bradycardia, **Free play** (starts in sinus rhythm; you change it), or **Custom scenario**.
4. Check age, weight and sex. A patient under 18 gets the paediatric hint cards, energies and drug prompts.
5. Press **Start Defib Sim**, then link the tablet with **Join** (QR code) or **Open defib**.

Custom scenarios run a sequence of up to five rhythms. Each step moves on at a trigger you pick: on analyse, on shock, on pacing capture, or after 30 s, 60 s or 2 min. Save, Load, Export file and Import file keep sequences on that device or share them. During the run, **Next step** and **Restart sequence** let you move it on by hand.

The clock. It starts when you press **START**, or by itself at the learner's first press on the defib. A banner warns you while it is not running. If you **PAUSE**, presses are still logged but the clock stays stopped until you press **START**.

The Defib controller

Area	What it does
Patient	Who the patient is, plus tags such as Needs SYNC, Needs pacing and the RCUK energy advice
What the pads show	The rhythm and the patient's true obs (the BP here is the live BP, not the last cuff reading)
The learner's defib	A live mirror of the tablet: mode, energy, SYNC, pacer settings and the screen message, plus shocks given
Rhythm	Click a rhythm to change it; All rhythms shows the full list

Area	What it does
Shock response (closed)	How shocks resolve, and Cardioversion succeeds at (see Arrests, rhythms and the defibrillator)
Pacing & artefacts (closed)	The capture threshold (± 5 mA; a pulse needs about 10 mA more), and movement, mains interference or lead-off artefact
Arrest	Time in arrest and shocks, the cycle timer, Start CPR, Metronome, Rhythm check (+2:00), ROSC, Capnography and Cycle NIBP . Capnography on puts a capnogram under the ECG and the ETCO2 number on the learner's defib; while it is off, the defib shows "--" for ETCO2, as a real one without its CO2 sensor does. The defib's NIBP shows only the last cuff reading: it changes when the learner presses NIBP on the defib or you press Cycle NIBP
RCUK prompts	When adrenaline or amiodarone is due. You see these in both modes; the learner never does
Drugs (closed)	Adrenaline, amiodarone, atropine, adenosine, isoprenaline, adrenaline infusion, fentanyl and midazolam
Event log	Every press on the defib with its time; add notes and flags

Closed sections show a one-line summary, for example "Never converts · cardioversion at 200 J".

How shocks resolve by default. Defib Sim uses **Auto**: an arrest converts on the third adequate shock, and a cardioversion on the first adequate synchronised shock. "Adequate" means at least 150 J for an adult (3 J/kg for a child) to defibrillate, and 70 J (1 J/kg) to cardiovert. These thresholds are simulator settings, not guideline values. An unsynchronised shock into a rhythm with a pulse causes VF.

Pacing captures electrically at the scenario's threshold (it varies by up to 15 mA each run) and gives a pulse about 10 mA above it. Demand pacing is inhibited by a faster native rate.

Debrief. The debrief lists good practice and areas for improvement: pulse checks, mode, SYNC, energies, time to first shock, drug timing, sedation before cardioversion, analgesia for pacing and capture. **Certificate** prints a certificate of completion.

Paediatric scenarios

Child scenarios follow the Resuscitation Council UK (RCUK) Guidelines 2025: the Paediatric advanced life support and Paediatric cardiac arrhythmias algorithms, and the Paediatric emergency drug chart.

WETFLAG. The brief for a child scenario shows the WETFLAG values worked out from the child's weight. If you do not give a weight, it is estimated from age using the RCUK chart.

WETFLAG starts hidden on the room monitor. To show it to the team, press **Show WETFLAG on Monitor** in the Rhythm & resus section; press **Hide WETFLAG on Monitor** to take it away again. To have it showing from the start, tick **Show WETFLAG on Monitor from the start** on the setup screen.

Letter	Value used
W	Weight in kg
E	Defibrillation energy, 4 J/kg
T	Tracheal tube size (cuffed and uncuffed), from the RCUK chart
F	Fluid bolus 10 ml/kg, up to 500 ml
L	Lorazepam 0.1 mg/kg, up to 4 mg
A	Adrenaline 10 micrograms/kg, up to 1 mg
G	10% glucose 2 ml/kg, up to 50 ml

Doses during the scenario. When the team gives a drug to a child, the log shows the weight-based dose (for example atropine, adenosine, amiodarone, buccal midazolam, levetiracetam, 10% glucose and calcium gluconate). IM adrenaline for anaphylaxis follows the age bands. **Tools → Drug calculator** works out doses for any weight.

Shocks for a child. Defibrillation is 4 J/kg. From the 5th shock in refractory VF or pulseless VT, up to 8 J/kg is accepted without a flag. Cardioversion starts at 1 J/kg and doubles up to 4 J/kg.

Newborns. Newborn scenarios show the Newborn life support steps (Guidelines 2025) as prompts in the log.

The event log, notes and flags

Everything that happens is logged with its scenario time: treatments, obs changes, rhythm changes, shocks, monitoring on or off, and your notes. The log feeds the debrief.

- **Where:** the Event log section at the bottom right of the instructor screen, newest first. It stays on screen throughout. **Full log** (or **Tools → Full simulation log**) opens the whole log.
- **Add a note:** type in the box under the log and press **Add note**, for example "Team leader identified".
- **Add a flagged note:** type the note and press **Flag** instead.
- **Flag an existing entry:** click the small flag beside it. Click again to remove the flag.

Flagged entries are listed under **Flagged events** in the debrief and marked on its graph, so you can find the moments to discuss.

Safety flags count sequence deviations: actions done before their usual prerequisite was in place, such as an IV drug before IV access. Nothing is blocked; the team can still do it. When there is at least one, an amber **Safety flags** button appears in the top bar. It opens the list, with arrests, shocks and your flagged moments shown separately as significant events. Safety flags are not used in Quick Sim.

Finishing and the debrief

Press **Finish** (or **F**) and confirm. The debrief opens on your laptop only. The room monitor shows "Simulation complete" and asks the team to turn to you, with no score or diagnosis.

Part of the debrief	What it shows
Performance Summary	Objectives met, the score (fully met, and with partial credit) and the time taken
Learning Objectives	Each objective as fully met, partly done or not in place, with the steps behind it
Vitals Trend & Interventions	A graph of the obs with each treatment marked; dashed lines mark flagged events
Flagged events	The log entries you flagged
Sequence Deviations	The safety flags
Defibrillation & Rhythm	Shocks, energies and each rhythm change
Defib Sim feedback	Defib Sim only: good practice and areas for improvement
Session Replay	Drag the slider to show the obs at any moment
Simulation Log	Every entry with its time
Instructor Notes	A box for your own feedback notes

Quick Sim has no scenario, so it has no score; the graph, log and replay still work.

Keeping a copy

- **Download Report** saves the debrief as a web page you can open on any computer, including the log and your instructor notes. Write your notes before you download.
- **Print** opens the report ready to print or save as PDF.
- **Certificate** (Defib Sim only) asks for the learner's name and prints a certificate of completion.

Exit to Menu returns to the setup screen. The session code and any linked screens stay the same, ready for the next scenario.

Tools, sound and shortcuts

The **Tools ▼** menu in the top bar holds the extras.

Tool	What it does
Drug calculator (D)	Doses for any weight you enter; child doses follow the RCUK chart
Timer alerts (T)	Set a reminder at a scenario time, for example "5 mins: bloods back". It plays a tone, appears in the log, and is marked FIRED
Full simulation log	The whole event log in a window
Sound plays on	Choose where the pulse tone, alarms, the cuff noise, sound effects and the patient's voice play: room monitor (the default), this controller, or both. Mute alarms silences the pulse tone and alarms only. The defib tablet always makes its own sounds (charging, shock, metronome and, in Defib Sim, the cuff)
Mute / Unmute alarms	While muted, a red Muted button stays in the top bar so alarms are never silently off
Keyboard shortcuts (?)	The list below

Room monitor timer. Show Timer, above the waveforms, puts a timer on the room monitor that you start and pause from your screen. It is useful for the team's own timings.

Keyboard shortcuts (they do nothing while a pop-up is open or a box or button is selected; if one does not work, click an empty part of the screen first)

Key	Action
Space	Pause or start the clock
F	Finish the scenario
D	Drug calculator
T	Timer alerts
?	Show the shortcuts

Troubleshooting

Problem	Likely cause	Fix
Room monitor shows no numbers or "No sensor detected"	Monitoring not attached yet	Press Attach , or click the chips in Monitoring & access
A value you set is not on the room monitor	That sensor is not attached (your tile says "not on monitor")	Attach it when the team does
BP on the room monitor has not changed	The cuff has not cycled (this is deliberate)	Press Cycle NIBP Now , or use Change and cycle cuff now when you set the BP
No sound	Sound not enabled, muted, or playing on the other screen	Tap Tap to Enable Sound on the monitor; check for the red Muted button; check Tools → Sound plays on ; check the device's volume
Badge says Sync error	The internet connection has dropped	Check the connection on each device
Room monitor or defib not following your changes	It is on a different session code, for example after New code	Join it again with the current code
Defib tablet asks for a code	It was opened without one	Type the session code, or scan the QR code from Join
Defib Sim log times all 00:00	The clock has not started	Press START ; it also starts at the learner's first press
Space bar or other shortcut does nothing	A button, box or pop-up is selected	Click an empty part of the screen and try again
Instructor tab closed or crashed	—	Reopen the site and press Resume under "Resume Previous?"
The Restricted tab is locked	Restricted scenarios need an account with access	Sign in, then use Request access
A screen looks out of date after an update	The browser is showing the old version	Reload the page